

Table 1: Product design specification for the aerosol extraction system prototype framed as Functions, Objectives, and Constraints (FOC) (extracted from final report).

Category	Requirement	Specification
Functions	Mist Extraction	Remove airborne low-inertia satellite droplets and ink mist from the printzone during operation via downstream suction.
	Selective Action	Preferentially entrain low-inertia satellite droplets while leaving the trajectories of the main printing droplets effectively unchanged.
	Continuous Operation	Operate continuously over typical industrial print durations without loss of basic functionality.
	Flow Regulation	Allow for the adjustment of suction pressure to tune for specific situations, accompanied by active differential pressure measurement.
	Condition Monitoring	Monitor ambient temperature and relative humidity for experimental characterisation, and provide indicators of degraded operating conditions (e.g., pressure changes due to filter loading).
Objectives	Extraction Efficiency	- Achieve a high capture rate of airborne mist-forming satellites, targeting a 50% capture rate corresponding to 20 g/s. - Actual performance can be gauged by the darkening of a HEPA filter.
	Suction Strength	Minimise pressure drop or losses across the system to ensure operating efficiency.
	Maintainability	Provide straightforward access for the cleaning of deposited ink and the rapid replacement of filters or internal flow components.
	Assembly Simplicity	Minimise the number of parts with interchangeable parts, enabling straightforward assembly and installation using simple fasteners (e.g., bolts).
	Manufacture Simplicity	The design should be easily manufactured using accessible methods like additive manufacturing, machining and standard workshop fabrication.
	Modularity	Ensure design elements (e.g., extractor geometry, electronics) are easily swappable and expandable to allow for rapid iteration and future improvements.

Table 1: Product design specification for the aerosol extraction system prototype framed as Functions, Objectives, and Constraints (continued).

Category	Requirement	Specification
Constraints	Geometric Limits	The system must conform to a maximum bounding box of 770 mm × 170 mm × 750 mm (free space available in printers).
	Spatial Compatibility	The extractor must maintain clearance for the printgap and absolutely must not obstruct printheads, sensors, maintenance access points, or other critical printer sub-systems.
	Capture Zone	The extractor inlet geometry must possess a sufficiently large capture zone of 350 mm in length to cover the working area of the printhead.
	Material Integrity	- Materials must be sufficiently stiff to maintain alignment and stability during operation. - Prolonged exposure to inks must not lead to significant chemical degradation (though superficial aesthetic staining is acceptable).
	Electrical Safety	Any electronics used for sensing or monitoring must operate safely at low voltages and be suitable for safe integration into a prototype system.